STEP 1 — Update system

sudo dnf update -y

STEP 2 — Install Apache (generate traffic)

sudo dnf install -y httpd
sudo systemctl enable httpd
sudo systemctl start httpd

Test:

http://WORKER-IP

STEP 3 — Install Node Exporter

Download

cd /opt
sudo curl -LO
https://github.com/prometheus/node_exporter/releases/download/v1.8.2/node_exporter-1.8.2.linux-amd64.tar.gz

---

Extract

sudo tar -xvf node_exporter-1.8.2.linux-amd64.tar.gz

Move binary

sudo mv node_exporter-1.8.2.linux-amd64/node_exporter /usr/local/bin/

STEP 4 — Create service

sudo vi /etc/systemd/system/node_exporter.service

Paste:

```
[Unit]
Description=Node Exporter
After=network.target

[Service]
User=ec2-user
ExecStart=/usr/local/bin/node_exporter

[Install]
WantedBy=default.target
```

STEP 5 — Start service

```
sudo systemctl daemon-reload
sudo systemctl enable node_exporter
sudo systemctl start node_exporter
```

STEP 6 — Verify

http://WORKER-IP:9100/metrics

⚙ 5. MASTER SETUP

STEP 1 — Install Prometheus

```
cd /opt
sudo curl -LO
https://github.com/prometheus/prometheus/releases/download/v2.52.0/prometheus-2.52.0.linux-amd64.tar.gz
```

STEP 2 — Extract

```
sudo tar -xvf prometheus-2.52.0.linux-amd64.tar.gz
sudo mv prometheus-2.52.0.linux-amd64 prometheus
```

STEP 3 — Create data folder

```
sudo mkdir -p /opt/prometheus/data
sudo chown -R ec2-user:ec2-user /opt/prometheus
```

STEP 4 — Configure Prometheus

sudo vi /opt/prometheus/prometheus.yml

Paste:

```
global:
  scrape_interval: 15s

scrape_configs:
  - job_name: "workers"
    static_configs:
      - targets:
          - "WORKER-PRIVATE-IP:9100"
```

STEP 5 — Create service

sudo vi /etc/systemd/system/prometheus.service

Paste:

```
[Unit]
Description=Prometheus
After=network.target

[Service]
User=ec2-user
ExecStart=/opt/prometheus/prometheus \
  --config.file=/opt/prometheus/prometheus.yml \
  --storage.tsdb.path=/opt/prometheus/data

[Install]
WantedBy=default.target
```

STEP 6 — Start Prometheus

```
sudo systemctl daemon-reload
sudo systemctl enable prometheus
sudo systemctl start prometheus
```

STEP 7 — Verify

http://MASTER-IP:9090

Check:

- Status → Targets → UP

📊 6. INSTALL GRAFANA (MASTER)

STEP 1

sudo dnf install -y https://dl.grafana.com/oss/release/grafana-11.0.0-1.x86_64.rpm

---

STEP 2

```
sudo systemctl enable grafana-server
sudo systemctl start grafana-server
```

STEP 3

Open:

http://MASTER-IP:3000

Login:

admin / admin

🔗 7. CONNECT GRAFANA → PROMETHEUS

Inside Grafana:

- Settings → Data Sources
- Add Prometheus

URL:

http://localhost:9090

Save & Test ✓

📊 8. IMPORT DASHBOARD

- Dashboards → Import
- Enter:

1860

Select Prometheus → Import

🔥 9. GENERATE LOAD (DEMO)

On worker:

```
sudo dnf install -y httpd-tools
ab -n 10000 -c 100 http://localhost/
```

🎯 FINAL RESULT

Students will see:

CPU ↑
Memory ↑
Network ↑
Load changes live

🧠 SIMPLE EXPLANATION FOR STUDENTS

Component	Role
Node Exporter	Collects system metrics
Prometheus	Stores metrics
Grafana	Shows dashboards

🚀 OPTIONAL (NEXT CLASS IDEAS)

If you want to extend teaching:

- Alerts (CPU > 80%)
- Multiple workers auto-setup script
- AWS EC2 auto-discovery
- Monitoring Apache (httpd metrics)
- Production architecture design